Aaghash R

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Professional Summary

Senior Software Engineer with 7+ years of experience in Robotics, Industrial Automation, Industrial Communication, and IIoT systems. Expertise in C, C++, Python, ROS, Embedded Systems, OPC UA, MQTT and Machine Vision with strong experience in scalable industrial software development, robotics integration, and communication frameworks.

Work Experience

ABB Robotics Bangalore, India

Senior Software Engineer – R&D Robotics Oct 2023 – Present

- Actively developing and integrating ROS2 interfaces into ABB's robotics software framework to enable seamless interoperability with robotic applications.
- Developing a ROS2 robot driver providing direct access to robot motion interfaces over OPC UA, enabling real-time communication and control.
- Architected the migration from a proprietary OPC UA stack to the open62541 SDK
- Own the investigation and resolution of customer-reported Salesforce issues across three industrial communication products, improving product reliability and customer support.
- Reduced production defects by **20%** through regression testing, root-cause analysis, and corrective actions.
- Increased team productivity by **30%** by driving Agile development practices and Scrum ceremonies.

DiFACTO Robotics and Automation Bangalore, India

Senior Engineer - Robotics Aug 2022 - Sep 2023

- Led development of robotics automation and machine vision applications for industrial manufacturing systems.
- Implemented vision-guided robotic correction systems for EV battery pack welding automation.
- Delivered automation solutions for international solar panel inspection systems deployed in Malaysia.

Development Engineer - Technology Development Dec 2020 - Jul 2022

- Developed vision software using C++, Python, OpenCV, and Qt for inspection, barcode detection, robot guidance, and analytics.
- Built OPC UA-based IIoT dashboards for real-time robot monitoring and production analytics.
- Designed ROS-to-PLC communication interfaces enabling AMR navigation and industrial integration.

Harita Seating Systems Ltd Hosur, India

Senior Engineer - Central Production Engineering Aug 2019 – Nov 2020

- Developed robotic automation solutions focused on manufacturing productivity improvement.
- Collaborated on autonomous system deployment and robotic integration projects.

Graduate Engineer Trainee - Central Production Engineering Aug 2018 – Jul 2019

- Built real-time robot productivity monitoring dashboards using SQL and Windows applications.
- Performed robot process analysis and optimization for production efficiency.

Professional Achievements

- Received a customer achievement award from **Fanuc** for delivering an automated generator rotor assembly handling solution supporting **200+ product variants**, achieving **0.1 mm positioning accuracy** using a dual-robot system integrated with machine vision.
- Conducted a Proof of Concept comparing a proprietary motion planning solution with **MoveIt2**, achieving a **15% improvement** in planning performance and validating it as a viable framework for robotic manipulation and path planning.
- Led the design and development of a novel robotics Proof of Concept currently under patent review, with technical details withheld due to confidentiality and intellectual property requirements.
- Reduced OPC UA licensing costs by **$130K** over three years by leading the migration from a proprietary OPC UA stack to open62541.
- Developed a scalable IIoT Gateway supporting **1,000+** concurrent industrial device connections with **99.9%** uptime using OPC UA and MQTT.

Skills

Programming Languages	C, C++, Python, Java, C#
Robotics & Middleware	ROS2, OPC UA, MQTT, Industrial Communication Protocols
Computer Vision & AI	OpenCV, Machine Vision, Vision-guided Robotics, Robot Perception
Industrial Robotics	Fanuc, ABB, KUKA, Epson, Motoman, Kawasaki
Industrial Applications	Pick and Place, Arc Welding, Spot Welding, Resistance Welding, Machine Tending, Gluing, Sealing, BIW Welding
Web Technologies	React, Node.js, REST APIs
Tools	Git, Docker, CMake, MQTT, Visual Studio, RobotStudio
Platforms	Linux, Windows, VxWorks, Raspberry Pi, NVIDIA Jetson
Domains	Robotics, Embedded Systems, Industrial Automation, IIoT, Real-time Systems

Certifications

Certified ScrumMaster – Scrum Alliance
Machine Learning – Stanford University – Coursera
Design Patterns – University of Alberta – Coursera
Object-Oriented Design – University of Alberta – Coursera
Software Architecture – University of Alberta – Coursera
Service-Oriented Architecture – University of Alberta – Coursera

Education

PSG College of Technology — Coimbatore, India
B.E Robotics and Automation — 2014 – 2018